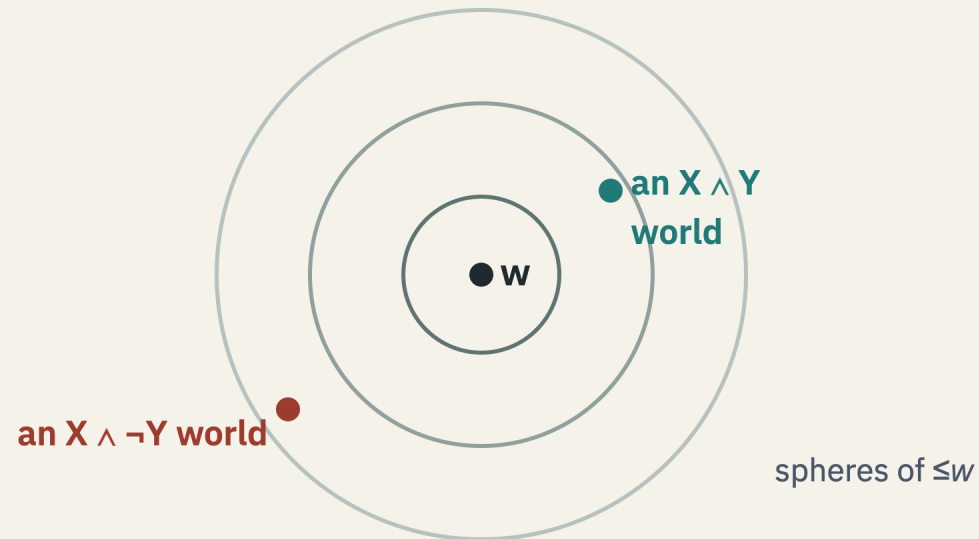


Counterfactuals Without the Ordering

Reopen, restate, re-prune: a semantics that never leaves the evaluator's own information state, and why its logic lies outside the preferential family.

The ordering is imposed, not derived

LEWIS, STALNAKER: TRAVEL TO THE CLOSEST WORLDS



$X \Box \rightarrow Y$ at w iff some $X \wedge Y$ -world is closer than any $X \wedge \neg Y$ -world, **by $\leq w$** , a parameter nothing in the picture derives.

THIS PAPER: OPERATE ON THE EVALUATOR'S OWN STATE

information state A

↓ reopen q_x : grounds cleared, the rest protected

blank intermediate

↓ restate $q_x = x$, then re-prune under the laws

pocket

pocket

pocket

the residue $\pi_x(A)$, weighted

$X \Box \rightarrow Y$ at A iff **every visible pocket affirms Y**. No world-similarity relation occurs on the right-hand side.

D1 clause-level deletion

theorem, by inspection

D2 qualitative non-recoverability

theorem: no smooth preferential relation agrees with it

D3 graded, indexed non-recoverability

open; the class to be named

The evaluation point is an information state, not a world

stated

an entry $q = v$ the state carries

blank

unresolved: no value stated, not necessarily every value open

closed

contains its own consequences under the laws

laws

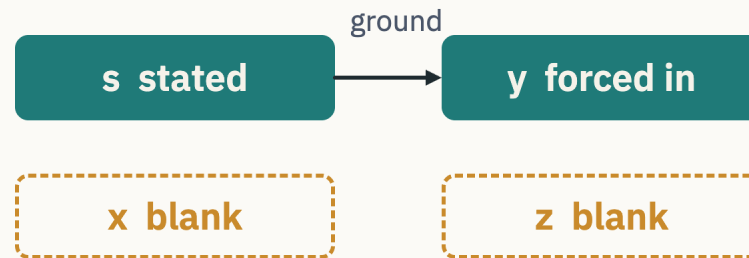
constraint-layer declarations, never entries, never retracted

declared support, Dep

grounds and dependents, declared; entailment is no part of it

constraint layer law $s \rightarrow y \cdot \{x, s, \neg z\}$ invalid · retained through every evaluation

INFORMATION STATE A

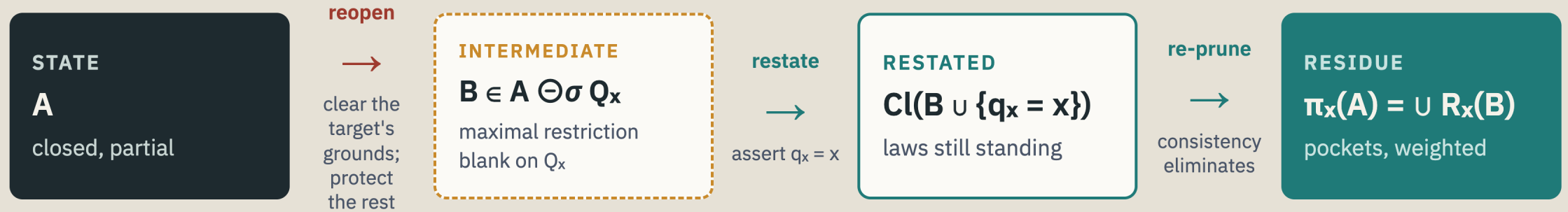


y is in the closure because s is stated and $s \rightarrow y$ is a law; s is y's sole declared ground. Blanking y therefore has to clear s, or closure puts y straight back.

INDEXED, NOT MENTAL

The index is any record-bearing structure, an instrument as readily as a person. Verdicts are facts about states, so differently informed evaluators disagree without contradiction. The past is as open as an unsettled future.

Reopen, restate, re-prune



ONE RESIDUE, THREE GRAINS

would every visible pocket affirms Y

might some visible pocket affirms Y

P(Y || X) weight of affirming pockets ÷ weight of all

THE SCOPE DIAL, Σ

evidential-minimal blank Q only

unidirectional Q and its declared dependents

bidirectional also its declared grounds

$R_x(B) = \text{MaxCons}\{P : C(B) \subseteq C(P), P \models X\}$. Laws over particulars; priorities antecedent-independent; infeasible where protected content and a law force an attribute of Q.

The witness: a held conjunct clears its ground

THE BOOKS, A

s the sprinklers ran *stated*

s $\rightarrow$ **y** a run leaves the lawn wet *law*; **s**
the sole ground of **y**

x it rained last night *blank*

{x, s, $\neg$ z} **invalid** rain on a run overflows
the gutters

z the gutters overflowed *blank*

had it rained (reopen **x**)

intermediate **{s, y}**: nothing cleared $\rightarrow$ **(x, s, y, z)** $\rightarrow$

lawn wet, gutters overflow: $x \vdash \sim y, x \vdash \sim z$

had it rained and the lawn been wet (reopen **x** and **y**, dual peg)

intermediate $\emptyset$: **y**'s ground **s** cleared $\rightarrow$ **(x, s, y, z)** **(x, $\neg$ s, y, z)** **(x, $\neg$ s, y, $\neg$ z)**

the sprinklers never ran in some pockets, and nothing overflows: $x \wedge y \not\vdash \sim z$

Boolean reading, the yes-peg of "is $x \wedge y$ true?": already blank, nothing cleared, one pocket,
the conclusion holds. The result is a result about the dual-peg reading.

Cautious Monotony fails at (x, y, z); Rational Monotony fails on the same instance, since $x \not\vdash \sim \neg y$. Every line is recomputed by the accompanying script, at all three scopes.

Not preferential: the Kraus–Lehmann–Magidor audit

Reflexivity

holds

Left Logical Equivalence

holds, by normalization

Right Weakening, And

hold outright

Or

holds: a disjunct's pockets are the disjunction's

Cut

holds when intermediates nest branchwise; can fail without nesting

Cautious Monotony

fails, by the witness

Rational Monotony

fails, same instance

Strong centering

fails, as a commitment: a stated toss is reopened like an open one

Where it fails

exactly when an added pocket omits cleared content and fails Z

THE CHAIN

preferential $\Rightarrow$ Cautious Monotony



ranked, hence Lewis and Stalnaker, $\Rightarrow$ preferential



no preferential relation on any superlanguage extends $\vdash \sim A$ (the witness is fully evaluable)



not representable by any smooth similarity ordering

Corroboration, not proof: 3,000 random finite models; the failure criterion agrees with the operator on all 50,291 nested singleton instances (Cautious Monotony failing in 3,113); no failure of Cut in 47,324 nested instances, of pocket monotonicity in 45,281, of Or in 53,986.

The same witness under the neighboring semantics

Semantics	$x \Box \rightarrow y$	$x \Box \rightarrow z$	$x \wedge y \Box \rightarrow z$	Cautious Monotony
This paper, dual-peg reading	yes	yes	no	fails
This paper, Boolean reading	yes	yes	yes	holds
Maximal-consistent-facts comparator (Lewis 1981)	yes	yes	yes	holds
Basis retraction, Veltman-type	no	no	no	not instantiated
AGM revision	yes	yes	yes	holds
Pearl, the structural model in the text	yes	yes	yes	holds
Update (Winslett)	yes	no	no	not instantiated

Only the dual-peg operator affirms both premises and rejects the conclusion.

For one identifiable reason: it clears the support of a reopened conjunct. Every other row keeps s.

Reopening is not settling.

Settling, as an intervention, keeps the causes standing and validates Cautious Monotony (composition). Reopening keeps every law and gives up the particular grounds. Two questions, two operations.

One residue, three grains; the indicative is a second operation

A RESIDUE $\Pi_X(A)$, WEIGHTS W

pocket affirms $Y \cdot 0.45$

affirms $Y \cdot 0.30$

$\neg Y \cdot 0.15$

0.10

would

false: one visible
pocket denies

might

true

$P(Y \parallel X)$

0.85

Empty residue: would vacuously true, might false, grade undefined. The kernel enters the grade alone.

subjunctive: reopen, restate, re-prune

"had Oswald not shot him, someone else would have": would false, might true at small positive weight, the grade near zero however the weight on Oswald runs

indicative: select among the standing pockets

"if Oswald didn't shoot him, someone else did": true at grade one; its grade is the conditional proportion, Adams's quantity

Imaging consumes the ordering; the grade does not.

Lewis's transport plan is the ordering applied pointwise; here mass follows dependency and validity through the blank intermediate. Verdicts are never entries, so the triviality results find no update dynamics to exploit.

The classic cases, as declared structure and scope

NIXON'S BUTTON

The button-to-launch law lives in the constraint layer, never retracted. The grim verdict needs no tuned weights, and no miracle inserts the antecedent.

BACKTRACKING

Reopen narrowly and the past stays fixed; let the clearing run into the antecedent's declared grounds and the backtracking reading appears. Two scope settings, no hidden metric.

MORGENBESSER'S COIN

Nothing wires the toss to the bet, so the heads entry stands in every pocket: "had you bet, you would have won" at grade one, at every setting.

THE STREETLIGHT

Evidential-minimal: infeasible, since "proceeded" is protected and excludes red under the law. Unidirectional: it clears as recorded posterity, and the car stops at grade one.

THE SINKHOLE

Decades of erosion were blank relative to every state before the collapse. The verdict of "had the light not been red" changes with the index, not with the time.

BIZET AND VERDI

The residue splits between both-French and both-Italian: both would be false, both might be true, the grades strictly between zero and one, with no ordering to adjudicate.

Antecedent strengthening, transitivity and contraposition are not secured by anything; a strengthened antecedent is a different perturbation, and the witnesses show the failures.

Proved, committed, open

PROVED

- D1: no world-similarity relation in the clauses
- D2: not preferential, on finite fully supported models under the dual-peg reading
- Cut under branchwise nesting; its failure without
- Or; where Cautious Monotony fails, exactly
- No preferential relation on any superlanguage extends $\vdash \sim A$

COMMITTED

- the dual-peg reading of "and" in an antecedent
- declared Dep, a scope family, antecedent-independent priorities
- a weight kernel, for the grades alone
- learning as transit, never overwrite: indices stand
- strong centering fails, and is meant to

OPEN

- D3: name the class outside D2 and test it on the graded, indexed verdicts
- the kernel's derivation; measure invariance
- completeness for the $\Box \rightarrow, \Diamond \rightarrow$ fragment
- priorities derived from closure
- embedded conditionals; the Kratzer-side flavors

The ordering's absence is a fact about the relation, not a choice of notation. The reading it rests on is testable: run the gutter sentence through the two-switches paradigm.